

SUMMARY

Full-stack software engineer focused on building reliable, well-tested applications with TypeScript, React, Node.js, and modern web tooling. Recent experience contributing to Swell, an open-source API testing platform, with work spanning security hardening, test infrastructure, Redux modernization, and frontend architecture. Previously co-founded and led engineering for a children’s coding education platform, building production systems across AWS, Node.js, TypeScript, and database infrastructure.

SKILLS

Languages: TypeScript, JavaScript (ES6), Python, C++
Frontend: React, Redux, Zustand, HTML5, CSS3/Sass, Tailwind CSS, Material UI, DaisyUI, Solid.js, React Router, TanStack Router, Vite, webpack
Backend & APIs: Node.js, Express, WebSocket, Socket.IO, REST APIs, JWT/Auth, Cloudinary, Arcjet
Desktop & Tooling: Electron, electron-builder, CLI tools, Git
Databases: PostgreSQL, MongoDB, Amazon Aurora
Cloud & DevOps: AWS EC2, AWS S3, AWS Elastic Beanstalk, AWS VPC, CloudFront, ELB/ALB, Docker, Kubernetes, GitHub Actions
Testing: Jest, Playwright, Mocha, Chai, Vitest

EXPERIENCE

Swell · Software Engineer

April 2023 – February 2025

- Implemented Content Security Policy (CSP) in Swell using nonce and SHA-256-based hashing to secure backend protocol testing, mitigate cross-site scripting (XSS) and code injection attacks, and protect user data without impacting application performance
- Reduced technical debt by refactoring legacy Redux architecture: migrated to Redux Toolkit, implemented type-safe hooks, expanded TypeScript coverage, removed deprecated functions and standardized state management to improve maintainability and code quality
- Developed a React-based binary file upload feature using Test-Driven Development (TDD) with Jest, enabling validation of backend server routes across different file types and payload sizes to expand API testing capabilities
- Increased overall test coverage by 15% by repairing failed tests and authoring new unit and integration tests using Jest, Mocha, Chai and Playwright; improved test reliability with Jest mocking and Chai semantic assertions
- Improved application reliability and stability by identifying and resolving security vulnerabilities in API Simulation Tool through root cause analysis and Node.js debugging, preventing crashes and runtime failures.
- Standardized CSS and Material UI theming across the application by unifying CSS classes, implementing Material UI theming architecture, and adding dark mode support to ensure a consistent and accessible user experience.

Mission Coders, LLC · CTO, Co-Founder

May 2018 – December 2022

- Transitioned from an on-premise monolithic server to a scalable AWS infrastructure with Elastic Load Balancing (ELB-ALB) and Elastic Compute Cloud (EC2), reducing resource overhead by 40%, enabling auto-scaling based on traffic patterns, and improving reliability and cost-efficiency across multiple regions
- Designed and implemented a scalable database solution using Amazon Aurora RDS to manage student and course registration; utilized Aurora’s auto-scaling capabilities and read replicas to accommodate fluctuating traffic and high read workloads and achieving 99% availability
- Led a six-member team to develop a children’s programming learning platform using JavaScript, Node.js, Express, TypeScript, and Redux; integrated Scratch to deliver an engaging, interactive educational tool, completing the project 5 weeks ahead of the deadline

Quark Virtual Services · Technical Manager, Co-Founder

April 2013 – October 2022

- Streamlined deployment for managed and unmanaged Virtual Private Servers (VPS) using KVM and OpenVZ; applied infrastructure-as-code principles and configuration management (Ansible, Puppet) to create reusable, version-controlled templates, reducing manual effort and overall deployment time by 15%
- Drove SEO strategy by conducting keyword research, optimizing on-page elements (title tags, meta descriptions, header tags, content), and established an off-page backlink campaign driving 35% organic traffic growth
- Increased conversion rate by 5% through SEO efforts by monitoring performance via Google Analytics, contributing to revenue growth and business objectives

PROJECTS

QuickChat – Real-time chat platform | github.com/howardsun-dev/quickchat

- Built a real-time messaging platform using React, Node.js/Express, MongoDB, and Socket.IO with authentication. Implemented Zustand for state management, optimistic UI updates, online/offline presence detection and secure media uploads via Cloudinary. Added request protection and rate limiting with Arcjet and built a responsive UI with Tailwind CSS and DaisyUI

Home Automation – Engineering Project

- Engineered a home automation system using Raspberry Pi and Arduino, programmed in C++ for low-level control and performance. Integrated subsystems like automatic blind control with precise timing and PWM. Optimized memory and code execution for smooth operation, increasing home energy efficiency by 10%

EDUCATION

Central Connecticut State University – B.S. Electronic Technology

2012

INTERESTS

Diving into Korean and Russian languages; geeking out over all things Sci-Fi, especially Dune or Star Trek; exploring new cities and embracing nature while camping